

Deepak Mallampati

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/Deepak0070 |
Kent, OH

PROFESSIONAL SUMMARY

Applied AI engineer shipping production LLM systems end to end: healthcare RAG (~35% retrieval precision, >90% grounded alignment), agentic workflows (>30% hallucination reduction), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Shipped Manga Lens solo on the Chrome Web Store with four AI vision providers under per-provider failure isolation. Master's in Computer Science, Kent State. Ready to ship production AI systems from day one. Targeting Forward Deployed Engineer roles at GenAI observability platforms: scale eval + grounding + hallucination measurement programs across enterprise AI teams; build custom integrations + MLOps pipelines on Python/TypeScript + Docker + public cloud; carry both engineering depth and customer-facing communication into multi-stakeholder rollouts.

CORE COMPETENCIES

GenAI Observability • Eval Harnesses • RAG + Grounding • Agentic Workflows • MLOps / LLMOps • Python / FastAPI • TypeScript / React • Docker / CI-CD • Multi-Provider LLM • Predictive ML

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval with an evaluation harness; recursive semantic chunking and transformer embeddings improved retrieval precision ~**35%**, cut irrelevant retrieval >**30%**, grounded response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability ~**25%**, hallucinations cut >**30%**.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation - direct observability primitives - cutting post-deploy defects ~**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall +**15-20%** on high-risk categories, precision held >**90%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability >**98%**, downstream model instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP; query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.
- Supported RBAC and audit logging for financial modules in a regulated environment.

*Machine Learning Engineer**Hyderabad, India - Nonprofit, Video Analytics, Applied NLP*

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations - agentic observability at the contract boundary. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction across four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama) with per-provider payload handling for worst-case failure isolation. 29 sites supported across Manifest V3 service workers, seven-day translation cache.

TypeScript, Manifest V3, Service Workers, Claude / GPT-4o / GPT-4.1 / Ollama

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt transformation, grounded image generation. Structured prompt-transformation layer drove ~30% contextual alignment gain and ~25-30% first-pass image success over naive direct-prompt orchestration.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

YOLOv8 Driver Drowsiness Detection

Real-time computer vision for driver drowsiness; unified YOLOv8 detection+classification cut inference latency ~30% over a 2-stage CNN baseline. Walk-forward validation and class-weighted training for imbalanced safety data.

PyTorch, YOLOv8, OpenCV, ONNX

EDUCATION

Master of Science, Computer Science - Kent State University*Aug 2023 - May 2025**Focus: Applied Machine Learning, NLP, Database Systems.***Bachelor of Technology, Computer Science - KL University***Jun 2019 - May 2023**Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).*

CERTIFICATIONS

*Deep Learning Specialization - DeepLearning.AI (Coursera)**2024*

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++

AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic (Claude), Hugging Face Transformers, Diffusers, YOLOv8

LLM / GenAI: RAG, agentic workflows, evaluation pipelines, grounding, hallucination measurement, structured outputs, prompt engineering, guardrails, embeddings, vector search

MLOps / Observability: FastAPI, Flask, REST, Docker, structured logging, load simulation, evaluation harness, Jenkins, GitHub Actions, Grafana

Frontend: React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

Data: PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy

Cloud / DevOps: Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda); ramping Kubernetes

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, GenAI Observability, Workflow Automation, Computer Vision, Multimodal